WORKED SOLUTION

STEP I 1997, Question 8

Pure • Roots of Exponential Equations

STEP I 1997, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q8

The key idea

A root of $x = a^x$ must be positive, since $a^x > 0$. That lets us take logarithms and turn the question into one about the zeros of $f(x) = \ln x - x \ln a$ on $x > 0$. For $a > 1$ this function rises and then falls, so it has zeros only if its maximum is at least 0. For $0 < a < 1$ it is strictly increasing, so it can cross zero at most once, and the question is whether it crosses at all.

Solution

Reducing to $f(x) = 0$

Throughout, $a > 0$, so $a^x = e^{x \ln a}$ is defined and positive for every real x. If $x \leq 0$ then $a^x > 0 \geq x$, so there are no roots with $x \leq 0$. For $x > 0$ both sides are positive, and since ln is strictly increasing,

$$x = a^x \iff \ln x = x \ln a \iff f(x) = 0, \qquad \text{where } f(x) = \ln x - x \ln a.$$

So the real roots of $x = a^x$ are exactly the zeros of f on $(0, \infty)$.

No roots when $a > e^{1/e}$

Here $a > e^{1/e} > 1$, so $\ln a > 0$. Differentiating,

$$f'(x) = \frac{1}{x} - \ln a,$$

which is positive for $0 < x < 1/\ln a$ and negative for $x > 1/\ln a$. So f increases and then decreases, and its maximum value on $(0, \infty)$ is

$$f\left(\frac{1}{\ln a}\right) = \ln\left(\frac{1}{\ln a}\right) - 1 = -\ln(\ln a) - 1.$$

Now $a > e^{1/e}$ gives $\ln a > 1/e$, and taking logarithms again (both sides positive), $\ln(\ln a) > \ln(1/e) = -1$. Hence

$$f(x) \leq -\ln(\ln a) - 1 < 0 \quad \text{for all } x > 0.$$

So f is never zero, and the equation $x = a^x$ has no real roots, as required.

Ask yourself What happens exactly at $a = e^{1/e}$, and for $1 < a < e^{1/e}$?

At $a = e^{1/e}$ the maximum is exactly 0, attained at $x = 1/\ln a = e$, so there is one (repeated) root $x = e$: the line $y = x$ is tangent to $y = a^x$ at (e, e). For $1 < a < e^{1/e}$ the maximum is positive, and $f \to -\infty$ at both ends, so there are two roots. (For $a = 1$ the only root is $x = 1$.) The question doesn't ask for these, but they show that $e^{1/e}$ is exactly the cut-off.

Exactly one root when $0 < a < 1$

Now $\ln a < 0$, so

$$f'(x) = \frac{1}{x} - \ln a = \frac{1}{x} + |\ln a| > 0 \quad \text{for all } x > 0,$$

and f is strictly increasing on $(0, \infty)$. Hence f has *at most one* zero.

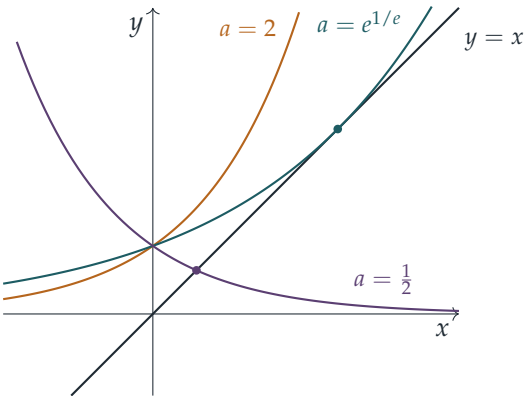
To see that it has one, note that f is continuous on $(0, \infty)$ and

$$f(1) = -\ln a > 0, \qquad f(a) = \ln a - a \ln a = (1 - a) \ln a < 0,$$

using $0 < a < 1$ in both. By the intermediate value theorem f has a zero between a and 1.

So for $0 < a < 1$ the equation $x = a^x$ has **exactly one** real root, and it lies in the interval $(a, 1)$.

The picture makes this clear: $y = a^x$ is decreasing and positive, while $y = x$ is increasing, so the two graphs cross exactly once. (Equivalently, $g(x) = a^x - x$ is strictly decreasing, with $g(0) = 1 > 0$ and $g(1) = a - 1 < 0$.)



Checks

- At the boundary, $(e^{1/e})^e = e$, so $x = e$ really is a root when $a = e^{1/e} \approx 1.4447$, matching the tangency.
- For $a = \frac{1}{2}$, bisection gives the single root $x \approx 0.6412$, which lies in $(\frac{1}{2}, 1)$ as predicted.
- Numerically, for 200 random $a > e^{1/e}$ the function $a^x - x$ stayed positive on a fine grid over $[-20, 20]$; the maximum of $\ln x - x \ln a$ matched $-1 - \ln \ln a$; for a just below $e^{1/e}$ there were two roots; and for 500 random $a \in (0, 1)$ there was exactly one sign change.

Common slips

- Taking logarithms without first noting that any root has $x > 0$: $\ln x$ is undefined otherwise, so the case $x \leq 0$ needs its own (one-line) argument.
- Showing only that f is increasing for $0 < a < 1$. That proves *at most* one root; you also need a sign change to show there is one.
- Getting the inequality the wrong way round when taking ln of $\ln a > 1/e$; it is safe because ln is increasing and both sides are positive.

Beyond STEP

The root for $0 < a < 1$ is a fixed point of the map $x \mapsto a^x$, and iterating $x_{n+1} = a^{x_n}$ leads to the "infinite power tower" $a^{a^{a^{\cdots}}}$. Euler showed that the tower converges exactly when $e^{-e} \leq a \leq e^{1/e}$, so the same cut-off $e^{1/e}$ appears again. Studying when iterations converge leads to the contraction mapping theorem of first-year analysis; Abbott, *Understanding Analysis*, covers the intermediate value theorem and the background needed.

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